

PGR106

Linear Algebra

Lecture 2 – Matrices



Matrices in the context of Data Science

Day	Maximum Temperature (° C)	Minimum Temperature (° C)	Snow (cm)	Dew (°)	Feels Like (° C)	Humidity (%)	Wind Speed (kmph)
Day-1	28	15	0	10	28	31	11
Day-2	28	16	0	12	28	36	12
Day-3	27	16	0	13	28	41	10
Day-4	28	16	0	11	28	35	10
Day-5	28	15	0	8	28	26	10
Day-6	25	18	0	17	25	68	20

Columns

Rows

[

28	15	0	10	28	31	11
28	16	0	12	28	36	12
27	16	0	13	28	41	10
28	16	0	11	28	35	10
28	15	0	8	28	26	10
29	17	0	11	30	45	13

]

Matrix

Matrices in the context of Data Science

		Max Temperature	Min Temperature	Snow	Dew	Feels Like	Humidity	Wind Speed
Day-1	→	28	15	0	10	28	31	11
Day-2	→	28	16	0	12	28	36	12
Day-3	→	27	16	0	13	28	41	10
Day-4	→	28	16	0	11	28	35	10
Day-5	→	28	15	0	8	28	26	10
Day-6	→	29	17	0	11	30	45	13

Matrix

Matrices in the context of Data Science

Variables / Attributes

Observations / Samples

		Max Temperature	Min Temperature	Snow	Dew	Feels Like	Humidity	Wind Speed
Day-1	→	28	15	0	10	28	31	11
Day-2	→	28	16	0	12	28	36	12
Day-3	→	27	16	0	13	28	41	10
Day-4	→	28	16	0	11	28	35	10
Day-5	→	28	15	0	8	28	26	10
Day-6	→	29	17	0	11	30	45	13

Matrix

Matrices in the context of Data Science

$$A = \begin{bmatrix} 28 & 15 & 0 & 10 & 28 & 31 & 11 \\ 28 & 16 & 0 & 12 & 28 & 36 & 12 \\ 27 & 16 & 0 & 13 & 28 & 41 & 10 \\ 28 & 16 & 0 & 11 & 28 & 35 & 10 \\ 28 & 15 & 0 & 8 & 28 & 26 & 10 \\ 29 & 17 & 0 & 11 & 30 & 45 & 13 \end{bmatrix}$$

Matrix

Elements of matrix A: $a_{i,j}$

row

column

Matrices in the context of Data Science

Matrix

$$A = \begin{bmatrix} 28 & 15 & 0 & 10 & 28 & 31 & 11 \\ 28 & 16 & 0 & 12 & 28 & 36 & 12 \\ 27 & 16 & 0 & 13 & 28 & 41 & 10 \\ 28 & 16 & 0 & 11 & 28 & 35 & 10 \\ 28 & 15 & 0 & 8 & 28 & 26 & 10 \\ 29 & 17 & 0 & 11 & 30 & 45 & 13 \end{bmatrix}$$

$\leftarrow a_{3,:}$

$\uparrow$
 $a_{:,5}$

Elements of matrix A : $a_{i,j}$

i^{th} row of matrix A : $a_{i,:}$

j^{th} column of matrix A : $a_{:,j}$

Dimension (Size)

- **Dimension**: is the number of rows and the number of columns
- **Square Matrix**: same number of rows and columns
- **Rectangular Matrix**: different number of rows and columns
 - rows < columns: it is called **Wide** Matrix
 - columns < rows: it is called **Tall** Matrix
- **Diagonal Elements**: elements with the same row and column index

Rectangular & Square Matrices

$$A = \begin{bmatrix} 1 & -3 & 4 \\ 3 & 6 & 2 \end{bmatrix}$$

$$2 \times 3$$

Rectangular Matrix

$$B = \begin{bmatrix} 1 & 4 \\ 3 & 5 \end{bmatrix}$$

$$2 \times 2$$

Square Matrix

$$C = \begin{bmatrix} 1 & -3 & 4 \\ 3 & 6 & 2 \\ -2 & 4 & 7 \end{bmatrix}$$

$$3 \times 3$$

Square Matrix

$$D = \begin{bmatrix} 1 & -3 \\ 4 & 3 \\ 6 & 2 \end{bmatrix}$$

$$3 \times 2$$

Rectangular Matrix

Wide & Tall Matrices

$$A = \begin{bmatrix} 1 & -3 & 4 \\ 3 & 6 & 2 \end{bmatrix}$$

$$2 \times 3$$

Rectangular Matrix

Wide Matrix

$$D = \begin{bmatrix} 1 & -3 \\ 4 & 3 \\ 6 & 2 \end{bmatrix}$$

$$3 \times 2$$

Rectangular Matrix

Tall Matrix

Diagonal Elements

$$A = \begin{bmatrix} 1 & -3 & 4 \\ 3 & 6 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 4 \\ 3 & 5 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & -3 & 4 \\ 3 & 6 & 2 \\ -2 & 4 & 7 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & -3 \\ 4 & 3 \\ 6 & 2 \end{bmatrix}$$

Matrix Transpose

Day	Maximum Temperature (° C)	Minimum Temperature (° C)	Snow (cm)	Dew (°)	Feels Like (° C)	Humidity (%)	Wind Speed (kmph)
Day-1	28	15	0	10	28	31	11
Day-2	28	16	0	12	28	36	12
Day-3	27	16	0	13	28	41	10
Day-4	28	16	0	11	28	35	10
Day-5	28	15	0	8	28	26	10
Day-6	25	18	0	17	25	68	20

Columns

Rows





28	15	0	10	28	31	11
28	16	0	12	28	36	12
27	16	0	13	28	41	10
28	16	0	11	28	35	10
28	15	0	8	28	26	10
29	17	0	11	30	45	13

Matrix

Matrix Transpose

		Variables / Attributes						
		Max Temperature	Min Temperature	Snow	Dew	Feels Like	Humidity	Wind Speed
Observations / Samples	Day-1 →	28	15	0	10	28	31	11
	Day-2 →	28	16	0	12	28	36	12
	Day-3 →	27	16	0	13	28	41	10
	Day-4 →	28	16	0	11	28	35	10
	Day-5 →	28	15	0	8	28	26	10
	Day-6 →	29	17	0	11	30	45	13

Matrix

Matrix Transpose

- **Transpose:** switch rows to columns and vice-versa

		Observations / Samples					
		Day-1	Day-2	Day-3	Day-4	Day-5	Day-6
Variables / Attributes	Max Temperature	28	28	27	28	28	29
	Min Temperature	15	16	16	16	15	17
	Snow	0	0	0	0	0	0
	Dew	10	12	13	11	8	11
	Feels Like	28	28	28	28	28	30
	Humidity	31	36	41	35	26	45
	Wind Speed	11	12	10	10	10	13
		Matrix					

Matrix Transpose

- **Transpose:** switch rows to columns and vice-versa (A^T)

$$A = \begin{bmatrix} 1 & -3 & 4 \\ 3 & 6 & 2 \end{bmatrix}$$

$$2 \times 3$$

$$A^T = \begin{bmatrix} 1 & 3 \\ -3 & 6 \\ 4 & 2 \end{bmatrix}$$

$$3 \times 2$$

$$A_{i,j} \longrightarrow A_{j,i}^T$$

Matrix Transpose

- **Transpose:** switch rows to columns and vice-versa ($\mathbf{B}^T$)

$$B = \begin{bmatrix} 1 & -3 & 4 \\ 3 & 6 & 2 \\ -2 & 4 & 7 \end{bmatrix}$$

3×3

$$B^T = \begin{bmatrix} 1 & 3 & -2 \\ -3 & 6 & 4 \\ 4 & 2 & 7 \end{bmatrix}$$

3×3

$$(B^T)^T \longrightarrow B$$

Symmetric Matrix

- Any matrix that is same as its **transpose** is known as **symmetric** matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n1} & a_{n2} & \cdot & \cdot & a_{nn} \end{bmatrix}$$

$$A = A^T$$

Symmetric Matrix

$$A = \begin{bmatrix} 1 & -3 & 4 \\ -3 & 6 & 2 \\ 4 & 2 & 7 \end{bmatrix}$$

$$3 \times 3$$

$$A^T = \begin{bmatrix} 1 & -3 & 4 \\ -3 & 6 & 2 \\ 4 & 2 & 7 \end{bmatrix}$$

$$3 \times 3$$

Matrix A is a Symmetric Matrix

$$A = A^T$$

Symmetric Matrix

	London	Paris	Amsterdam	Rome	Berlin
London	0	414	494	1799	1059
Paris	414	0	515	1531	1094
Amsterdam	494	515	0	1835	649
Rome	1799	1531	1835	0	1573
Berlin	1059	1094	649	1573	0

Distance Matrix

Symmetric Matrix

$$B = \begin{bmatrix} 1 & -3 & 4 \\ 3 & 6 & 2 \\ -2 & 4 & 7 \end{bmatrix}$$

$$3 \times 3$$

$$B^T = \begin{bmatrix} 1 & 3 & -2 \\ -3 & 6 & 4 \\ 4 & 2 & 7 \end{bmatrix}$$

$$3 \times 3$$

Matrix B is not a Symmetric Matrix

$$B \neq B^T$$

Symmetric Matrix

$$C = \begin{bmatrix} 1 & -3 & 4 \\ 3 & 6 & 2 \end{bmatrix}$$

$$2 \times 3$$

$$C^T = \begin{bmatrix} 1 & 3 \\ -3 & 6 \\ 4 & 2 \end{bmatrix}$$

$$3 \times 2$$

Matrix C is not a Symmetric Matrix

Symmetric Matrix

- A **symmetric** matrix is always a **square matrix**.
- A rectangular matrix **cannot** be a symmetric matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n1} & a_{n2} & \cdot & \cdot & a_{nn} \end{bmatrix}$$

$n \times n$

Identity Matrix

- **Identity** matrix: A square matrix where **diagonal elements** are **1** and all other elements are **0**

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

2×2
Identity Matrix

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3×3
Identity Matrix

$$I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4×4
Identity Matrix

Identity Matrix

- **Identity** matrix: A square matrix where **diagonal elements** are **1** and all other elements are **0**

$$I = \begin{bmatrix} 1 & 0 & 0 & . & . & 0 \\ 0 & 1 & 0 & . & . & 0 \\ 0 & 0 & 1 & . & . & 0 \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ 0 & 0 & 0 & . & . & 1 \end{bmatrix}$$

$n \times n$
Identity Matrix

Diagonal Matrix

- **Diagonal** matrix: A square matrix where **diagonal elements** are **non-zero** and all other elements are **0**

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$$

2×2

Diagonal Matrix

$$A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

3×3

Diagonal Matrix

$$A = \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 7 \end{bmatrix}$$

4×4

Diagonal Matrix

Diagonal Matrix

- **Diagonal** matrix: A square matrix where **diagonal elements** are **non-zero** and all other elements are **0**

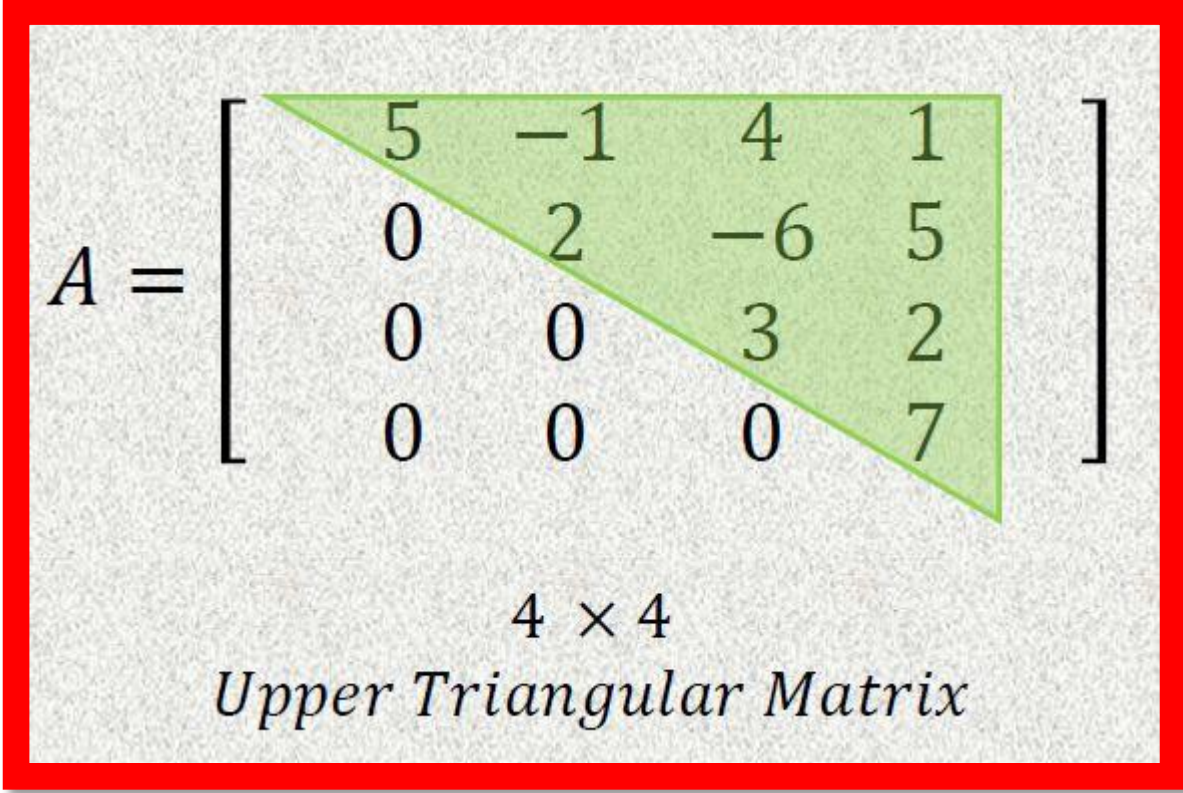
$$A = \begin{bmatrix} a_{1,1} & 0 & 0 & \cdot & \cdot & 0 \\ 0 & a_{2,2} & 0 & \cdot & \cdot & 0 \\ 0 & 0 & a_{3,3} & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot & a_{n,n} \end{bmatrix}$$

$n \times n$
Diagonal Matrix

Triangular Matrix

- A matrix is an **upper triangular** matrix if

$$A_{i,j} = 0 \text{ for } i > j$$



The matrix A is shown with its elements. The upper triangle, including the diagonal, is shaded green. The elements are:

$$A = \begin{bmatrix} 5 & -1 & 4 & 1 \\ 0 & 2 & -6 & 5 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 7 \end{bmatrix}$$

4 × 4
Upper Triangular Matrix

Triangular Matrix

- A matrix is a **lower triangular** matrix if

$$A_{i,j} = 0 \text{ for } i < j$$

$$A = \begin{bmatrix} 5 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 4 & -6 & 3 & 0 \\ 1 & 5 & 2 & 7 \end{bmatrix}$$

4×4

Lower Triangular Matrix

Zeros Matrix

- A **zeros** matrix is a matrix where all the elements are 0

$$0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

2×2
Zero Matrix

$$0 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

2×5
Zero Matrix

$$0 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

3×3
Zero Matrix

Zeros Matrix

- A **zeros** matrix is a matrix where all the elements are 0

$$0 = \begin{bmatrix} 0 & 0 & 0 & . & . & 0 \\ 0 & 0 & 0 & . & . & 0 \\ 0 & 0 & 0 & . & . & 0 \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ 0 & 0 & 0 & . & . & 0 \end{bmatrix}$$

$n \times m$
Zero Matrix

Ones Matrix

- A **ones** matrix is a matrix where all the elements are **1**

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

2×2
Ones Matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

2×5
Ones Matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

3×3
Ones Matrix

Ones Matrix

- A **ones** matrix is a matrix where all the elements are **1**

$$A = \begin{bmatrix} 1 & 1 & 1 & . & . & 1 \\ 1 & 1 & 1 & . & . & 1 \\ 1 & 1 & 1 & . & . & 1 \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ 1 & 1 & 1 & . & . & 1 \end{bmatrix}$$

$n \times m$
Ones Matrix

Matrix Addition

- **Matrix Addition** is performed by adding the corresponding elements in rows and columns.
- To perform matrix **addition**, matrices must be of the **same dimensions**.

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \quad B = \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{bmatrix}$$

$2 \times 2 \qquad \qquad \qquad 2 \times 2$

$$A + B = \begin{bmatrix} a_{1,1} + b_{1,1} & a_{1,2} + b_{1,2} \\ a_{2,1} + b_{2,1} & a_{2,2} + b_{2,2} \end{bmatrix}$$

2×2

Matrix Addition

Example

$$A = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \end{bmatrix}$$

$$2 \times 3$$

$$B = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 5 \end{bmatrix}$$

$$2 \times 3$$

$$A + B = \begin{bmatrix} 2 + 1 & 3 + 2 & 4 + 4 \\ -1 + 0 & 5 + 1 & -3 + 5 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 3 & 5 & 8 \\ -1 & 6 & 2 \end{bmatrix}$$

$$2 \times 3$$

Scalar Multiplication

- **Scalar Multiplication** is performed by multiplying each element of the matrix by a scalar.

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix}$$

2×2

$$2A = \begin{bmatrix} 2 * a_{1,1} & 2 * a_{1,2} \\ 2 * a_{2,1} & 2 * a_{2,2} \end{bmatrix}$$

2×2

Example

$$A = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \end{bmatrix}$$

2×3

$$2A = \begin{bmatrix} 2 * 2 & 2 * 3 & 2 * 4 \\ 2 * -1 & 2 * 5 & 2 * -3 \end{bmatrix}$$
$$2A = \begin{bmatrix} 4 & 6 & 8 \\ -2 & 10 & -6 \end{bmatrix}$$

2×3

Hadamard Product

- **Hadamard Product** (element-wise product) is performed by multiplying the corresponding elements of the matrix.
- To perform Hadamard product, matrices must be of the same dimensions.

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \quad 2 \times 2 \qquad B = \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{bmatrix} \quad 2 \times 2$$

$$A \odot B = \begin{bmatrix} a_{1,1} * b_{1,1} & a_{1,2} * b_{1,2} \\ a_{2,1} * b_{2,1} & a_{2,2} * b_{2,2} \end{bmatrix} \quad 2 \times 2$$

Hadamard Product

Example

$$A = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \end{bmatrix}$$

$$2 \times 3$$

$$B = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 5 \end{bmatrix}$$

$$2 \times 3$$

$$A \odot B = \begin{bmatrix} 2 * 1 & 3 * 2 & 4 * 4 \\ -1 * 0 & 5 * 1 & -3 * 5 \end{bmatrix}$$

$$A \odot B = \begin{bmatrix} 2 & 6 & 16 \\ 0 & 5 & -15 \end{bmatrix}$$

$$2 \times 3$$

Trace of a Matrix

- **Trace** of a matrix is defined as the sum of **all** the **diagonal elements** of the matrix.

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix}$$

$$2 \times 2$$

$$Tr(A) = a_{1,1} + a_{2,2}$$

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$$

$$3 \times 3$$

$$Tr(A) = a_{1,1} + a_{2,2} + a_{3,3}$$

Trace of a Matrix

- The **trace** operation is useful in verifying the **Eigenvalues** of the matrix.

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{bmatrix}$$

2×3

$$\text{Tr}(A) = a_{1,1} + a_{2,2}$$

$$A = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix}$$

3×3

$$\begin{aligned} \text{Tr}(A) &= 2 + 5 + 8 \\ &= 15 \end{aligned}$$

Matrix Multiplication

- Matrix Addition and Scalar Multiplication are performed element-wise.
- Hadamard Product operation is also element-wise.
- Matrix Multiplication is not element-wise.

Matrix Multiplication

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{bmatrix} \qquad B = \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \\ b_{3,1} & b_{3,2} \end{bmatrix}$$

2×3

3×2



Elements of AB → dot product (row of A and column of B)

$$AB = \begin{bmatrix} a_{1,1} * b_{1,1} + a_{1,2} * b_{2,1} + a_{1,3} * b_{3,1} & a_{1,1} * b_{1,2} + a_{1,2} * b_{2,2} + a_{1,3} * b_{3,2} \\ a_{2,1} * b_{1,1} + a_{2,2} * b_{2,1} + a_{2,3} * b_{3,1} & a_{2,1} * b_{1,2} + a_{2,2} * b_{2,2} + a_{2,3} * b_{3,2} \end{bmatrix}$$

2×2

Matrix Multiplication

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1p} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2p} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{m1} & a_{m2} & \cdot & \cdot & a_{mp} \end{bmatrix}$$

$m \times p$

$$B = \begin{bmatrix} b_{11} & b_{12} & \cdot & \cdot & b_{1n} \\ b_{21} & b_{22} & \cdot & \cdot & b_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ b_{p1} & b_{p2} & \cdot & \cdot & b_{pn} \end{bmatrix}$$

$p \times n$

$$AB = \sum_{k=1}^p A_{ik} B_{kj}$$

*column of Matrix A
or row of Matrix B*

*row
of
Matrix A*

*column
of
Matrix B*

Matrix Multiplication

Example

$$A = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 4 & 5 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 * 1 + 3 * 0 + 4 * 4 & 2 * 2 + 3 * 1 + 4 * 5 \\ -1 * 1 + 5 * 0 + -3 * 4 & -1 * 2 + 5 * 1 + -3 * 5 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 + 0 + 16 & 4 + 3 + 20 \\ -1 + 0 - 12 & -2 + 5 - 15 \end{bmatrix} = \begin{bmatrix} 18 & 27 \\ -13 & -12 \end{bmatrix}$$

Matrix Multiplication

Example

$$A = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 5 \\ -2 & -4 & -3 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 * 1 + 3 * 0 + 4 * -2 & 2 * 2 + 3 * 1 + 4 * -4 & 2 * 4 + 3 * 5 + 4 * -3 \\ -1 * 1 + 5 * 0 + (-3) * -2 & -1 * 2 + 5 * 1 + (-3) * -4 & -1 * 4 + 5 * 5 + (-3) * -3 \\ 6 * 1 + (-2) * 0 + 8 * -2 & 6 * 2 + (-2) * 1 + 8 * -4 & 6 * 4 + (-2) * 5 + 8 * -3 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 + 0 - 8 & 4 + 3 - 16 & 8 + 15 - 12 \\ -1 + 0 + 6 & -2 + 5 + 12 & -4 + 25 + 9 \\ 6 + 0 - 16 & 12 - 2 - 32 & 24 - 10 - 24 \end{bmatrix} = \begin{bmatrix} -6 & -9 & 11 \\ 5 & 15 & 30 \\ -10 & -22 & -10 \end{bmatrix}$$

Properties of Matrix Operations

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

$$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$$

$$s(\mathbf{A} + \mathbf{B}) = s\mathbf{A} + s\mathbf{B}$$

Properties of Matrix Operations

$$AB \neq BA$$



$$AI = IA = A$$

$$A(B + C) = AB + AC$$

$$(B + C)A = BA + CA$$

$$A(BC) = (AB)C$$

Properties of Matrix Operations

$$(A^T)^T = A$$

$$A^T B = B^T A$$

$$(A + B)^T = A^T + B^T$$

$$(AB)^T = B^T A^T$$

Matrix Power

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n1} & a_{n2} & \cdot & \cdot & a_{nn} \end{bmatrix}$$

$n \times n$

$$A^k = \underbrace{AA \dots \dots \dots A}_{k \text{ times}}$$

k times

Matrix Power

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$
$$2 \times 2$$

$$A^3 = AAA = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$A \rightarrow 2 \times 2 \text{ matrix}$

$A^k \rightarrow 2 \times 2 \text{ matrix}$

$A \rightarrow n \times n \text{ matrix}$

$A^k \rightarrow n \times n \text{ matrix}$

Diagonal Matrix Multiplication

$$\begin{matrix} 3 \times 3 \\ A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} \end{matrix} \quad \begin{matrix} 3 \times 3 \\ B = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix} \end{matrix}$$

$$AB = \begin{bmatrix} 2 * 2 + 0 * -1 + 0 * 6 & 2 * 3 + 0 * 5 + 0 * -2 & 2 * 4 + 0 * -3 + 0 * 8 \\ 0 * 2 + 3 * -1 + 0 * 6 & 0 * 3 + 3 * 5 + 0 * -2 & 0 * 4 + 3 * -3 + 0 * 8 \\ 0 * 2 + 0 * -1 + 5 * 6 & 0 * 3 + 0 * 5 + 5 * -2 & 0 * 4 + 0 * -3 + 5 * 8 \end{bmatrix}$$

$$AB = \begin{bmatrix} 4 + 0 + 0 & 6 + 0 + 0 & 8 + 0 + 0 \\ 0 - 3 + 0 & 0 + 15 + 0 & 0 - 9 + 0 \\ 0 + 0 + 30 & 0 + 0 - 10 & 0 + 0 + 40 \end{bmatrix} = \begin{bmatrix} 4 & 6 & 8 \\ -3 & 15 & -9 \\ 30 & -10 & 40 \end{bmatrix}$$

3×3

Diagonal Matrix Multiplication

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

3×3

$$B = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix}$$

3×3

$$AB = \begin{bmatrix} 4 & 6 & 8 \\ -3 & 15 & -9 \\ 30 & -10 & 40 \end{bmatrix}$$

3×3

← Scaled by 2

← Scaled by 3

← Scaled by 5

Diagonal Matrix Multiplication

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 * 2 + 3 * 0 + 4 * 0 & 2 * 0 + 3 * 3 + 4 * 0 & 2 * 0 + 3 * 0 + 4 * 5 \\ -1 * 2 + 5 * 0 + (-3) * 0 & -1 * 0 + 5 * 3 + (-3) * 0 & -1 * 0 + 5 * 0 + (-3) * 5 \\ 6 * 2 + (-2) * 0 + 8 * 0 & 6 * 0 + (-2) * 3 + 8 * 0 & 6 * 0 + (-2) * 0 + 8 * 5 \end{bmatrix}$$

$$BA = \begin{bmatrix} 4 + 0 + 0 & 0 + 9 + 0 & 0 + 0 + 20 \\ -2 + 0 + 0 & 0 + 15 + 0 & 0 + 0 - 15 \\ 12 + 0 + 0 & 0 - 6 + 0 & 0 + 0 + 40 \end{bmatrix} = \begin{bmatrix} 4 & 9 & 20 \\ -2 & 15 & -15 \\ 12 & -6 & 40 \end{bmatrix}$$

End of Lecture 2



MATHEMATICS FOR MACHINE LEARNING

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